## 12.9.3 Functions of Chloroplast

1. Photosynthesis 2. Storage of starch 3. Synthesis of fatty acids 4. Storage of lipids 5. Formation of chloroplasts

## 12.9.4 Photosynthesis

Photosynthesis (Photo = light; synthesis = to build) is a process by which autotrophic organisms like green plants, algae and chlorophyll containing bacteria utilize the energy from sunlight to synthesize their own food. In this process, carbon dioxide combines with water in the presence of sunlight and chlorophyll to form carbohydrates. During this process oxygen is released as a byproduct.




$$6CO_2 + 12H_2O \xrightarrow[\text{chlorophyll}]{\text{Light}} C_6H_{12}O_6 + 6H_2O + 6O_2\uparrow$$

Carbon dioxide + Water ⟶ Glucose + Water + Oxygen

## 12.9.5 Where does photosynthesis occur?

Photosynthesis occurs in all green parts of the plant especially in green leaves.

## 12.9.6 Photosynthetic Pigments

Pigments involved in photosynthesis are called **Photosynthetic pigments.** Photosynthetic pigments are of two classes namely, the primary pigments and accessory pigments. Chlorophyll a is the **primary pigment** that traps solar energy and converts it into electrical and chemical energy. Thus it is called the reaction centre. Other pigments such as chlorophyll b and carotenoids are called **accessory pigments** as they pass on the absorbed energy to chlorophyll a (Chl.a) molecule. Reaction centre (Primary pigments) and harvesting centre (Accessary pigments) together form photo systems.

## 12.9.7 Role of Sunlight in Photosynthesis

The entire process of photosynthesis takes place inside the chloroplast. The structure of chloroplast is such that the light dependent (**Light reaction**) and light independent (**Dark reaction**) take place at different sites in the organelle

**1. Light dependent Reaction (Hill reaction \ Light reaction)**

This was discovered by **Robin Hill** (1939). This reaction takes place in the presence of light energy in **thylakoid membranes** (grana) of the chloroplasts. Photosynthetic pigments absorb the light energy and convert it into chemical energy ATP and $NADPH+H^+$. These products of light reaction move out from the thylakoid to the stroma of the chloroplast.

### More to Know

| | |
|---|---|
| ATP | Adenosine Tri Phosphate |
| ADP | Adenosine Di Phosphate |
| NAD | Nicotinamide Adenine Dinucleotide |
| NADP | Nicotinamide Adenine Dinucleotide Phosphate |




A cell cannot get its energy directly from glucose. So in respiration the energy released from glucose is used to make ATP (Adenosine Triphosphate)

**2. Light independent reactions (Dark reaction) (Biosynthetic phase)**

Dark reaction or biosynthetic pathway is takes place in **stroma**. During this reaction $CO_2$ is reduced into carbohydrates with the help of light generated **ATP** and **$NADPH+H^+$**. This is also called as **Calvin cycle** and is carried out in the absence of light. It is called dark reaction.



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